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Chapter 4 – Automotive Hardware Security

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Summary

Comparison of Hardware Trust Anchors

	evita *full*	evita *medium*	evita *light*	HIS **SHE**	TRUSTED COMPUTING GROUP **TPM 2.0**	**Smartcard**
Cryptographic algorithms						
ECC/RSA	●/●	●/●	○/○	○/○	●/●	⊙/⊙
AES/DES	●/⊙	●/⊙	●/○	●/○	⊙/○	⊙/⊙
WHIRLPOOL/SHAx	●/●	●/●	○/○	○/○	○/●	⊙/⊙
Hardware acceleration						
ECC/RSA	●/○	○/○	○/○	○/○	○/○	○/○
AES/DES	●/○	●/○	●/○	●/○	○/○	○/○
WHIRLPOOL/SHAx	●/○	○/○	○/○	○/○	○/○	○/○
Security features						
Secure/authenticated boot	●/●	●/●	⊙/⊙	●/○	⊙/●	○/○
Key control per use/bootstrap	●/●	●/●	●/⊙	○/●	⊙/●	○/○
PRNG with TRNG seed	●	●	●	●	●	●
Monotonic counters 32/64 bit	●/●	●/●	●/●	○/○	●/●	○/○
Tick/UTC-synced internal clock	●/●	●/●	●/●	○/○	●/●	○/○
Internal processing						
Programmable/preset CPU	●/⊙	●/⊙	○/⊙	○/●	○/●	⊙/⊙
Internal V/NV (key) memory	●/●	●/●	⊙/⊙	●/●	●/●	●/○
Asynchronous/parallel IF	●/⊙	●/○	●/○	●/○	●/○	○/○

Annotation: ● = available, ○ = not available, ⊙ = partly or optionally available

Summary

- Hardware Trust Anchors are essential building blocks for automotive security
- Automotive HSMs
  - In addition, often provide cryptographic acceleration
  - Different types of HSMs, cf. EVITA small, medium, and full
  - SoC Integration
- (Automotive) TPM 2.0
  - Can be used in ECUs with high security requirements
    - High-end tamper resistance
    - Common Criteria EAL4+ certification
  - Complements automotive HSMs in other ECUs
- Automotive industry also uses other hardware security measures, e.g.,
  - Memory Protection Unit (MPU)
  - One Time Programmable Memory (OTP)

Literature

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Thank You

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